

Exp No:17 Experiment to understand Linear Regression for a given data set.

Description: Understand the Linear regression for the dataset given.

```
In [ ]: import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
```

```
In [19]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   YearsExperience  30 non-null    float64
 1   Salary          30 non-null    int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
In [3]: df.dropna(inplace=True)
```

```
In [4]: df.info()

<class 'pandas.core.frame.DataFrame'>
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Data columns (total 2 columns):
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---  -
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```

```
In [5]: df.describe()
```

```
Out[5]:
```

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

```
In [6]: features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
```

```
In [7]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_st
```

```
In [20]: from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
```

```
Out[20]:
```

LinearRegression

LinearRegression()

```
In [21]: model.score(x_train,y_train)
```

```
Out[21]: 0.9603182547438908
```

```
In [23]: model.score(x_test,y_test)
```

```
Out[23]: 0.9184170849214232
```

```
In [24]: model.coef_
```

```
Out[24]: array([[9281.30847068]])
```

```
In [25]: model.intercept_
```

```
Out[25]: array([27166.73682891])
```

```
In [26]: import pickle  
pickle.dump(model,open('SalaryPred.model','wb'))
```

```
In [27]: model=pickle.load(open('SalaryPred.model','rb'))
```

```
In [28]: yr_of_exp=float(input("Enter Years of Experience: "))  
yr_of_exp_NP=np.array([[yr_of_exp]])  
Salary=model.predict(yr_of_exp_NP)
```

```
Enter Years of Experience: 44
```

```
In [ ]:
```

```
In [29]: print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary)
```

```
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```

```
In [ ]:
```